

DELHI PUBLIC SCHOOL

Bokaro Steel City, Jharkhand

UNIT ASSESSMENT: PHYSICS

SUBJECT: PHYSICS

TIME ALLOWED: 2 HOURS

MAX MARKS: 60

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. Write your name, roll number, and section clearly in the provided space.
2. All questions are compulsory.
3. Use of calculators is permitted where necessary.
4. Review all answers before final submission.

SECTION A - MULTIPLE CHOICE QUESTIONS

Attempt all questions. Each question carries 1 mark(s).

- Q1. Which of the following is the correct unit of electric current? (1)
- (a) Ampere (A) (b) Volt (V)
(c) Ohm (Ω) (d) Watt (W)
- Q2. What happens to the resistance of a metallic conductor when its temperature is increased? (1)
- (a) It increases (b) It decreases
(c) It remains constant (d) It becomes zero
- Q3. Three resistors of 3Ω each are connected in parallel. What is the equivalent resistance of the circuit? (1)
- (a) 1Ω (b) 3Ω
(c) 9Ω (d) 0.33Ω
- Q4. The mathematical expression of Ohm's Law is: (1)
- (a) $V = IR$ (b) $I = VR$
(c) $R = VI$ (d) $P = VI$

SECTION B - SHORT QUESTIONS

Attempt all questions. Each question carries 2 mark(s).

- Q1. State Ohm's Law. Mention the conditions under which it holds true. (2)
- Q2. Define electric potential and potential difference. Write their SI units. (2)
- Q3. Why are electrical heating elements (like in bread toasters or hair dryers) made of an alloy rather than a pure metal? (2)

SECTION C - DIAGRAM/GRAPH-BASED QUESTIONS

Attempt all questions. Each question carries 5 mark(s).

- Q1. Draw a circuit diagram representing three resistors in parallel connected to a 12V battery, an ammeter, a voltmeter across one resistor, and a closed key. (5)
- Q2. Below is a V-I graph plotted for two different metallic wires, A and B. Wire A is steeper than wire B. Which of the wires has higher resistance? Justify your answer. (5)
- Q3. Identify and draw the standard symbols for: (a) A rheostat, (b) A plug key open, (c) A wire crossing without joining, (d) A light emitting diode (LED). (5)
- Q4. Draw the V-I characteristic curve for a non-ohmic conductor (like a vacuum diode or semiconductor diode) and explain why it does not follow a straight line. (5)
- Q5. Draw a circuit diagram representing three resistors in parallel connected to a 12V battery, an ammeter, a voltmeter across one resistor, and a closed key. (5)

SECTION D - NUMERICAL PROBLEMS

Attempt all questions. Each question carries 5 mark(s).

- Q1. An electric bulb is connected to a 220V generator. If the current drawn by the bulb is 0.50A, calculate the power of the bulb and its resistance. (5)
- Q2. An electric iron consumes energy at a rate of 840W when heating is at the maximum rate. The voltage is 220V. Calculate the current and the resistance. (5)
- Q3. A 100W electric bulb is lighted for 8 hours daily. Calculate the energy consumed in kilowatt-hours (kWh) by the bulb in a month of 30 days. (5)
- Q4. Find the equivalent resistance when two resistors of 10Ω and 40Ω are connected in parallel, and this combination is further connected in series with a 12Ω resistor. (5)
- Q5. A charge of 120 Coulombs flows through a bulb in 2 minutes. Find the magnitude of electric current passing through the circuit. (5)

